

Balancing efficiency and equity in location-allocation models with an application to strategic EMS design

S. Khodaparasti¹ · H. R. Maleki¹ · M. E. Bruni² ·
S. Jahedi¹ · P. Beraldi² · D. Conforti²

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Abstract This paper presents an integrated location-allocation model balancing efficiency and equity criteria. The new formulation combines two domains: facility location and data envelopment analysis. To support the decision maker with more realistic solutions based on the optimal location-allocation decisions, we endogenize the outputs of the model as a function dependent on the allocation variables. To illustrate the viability of the proposed approach, we investigated the potential application of the model to the design of an emergency medical service system.

Keywords Location-allocation problem · Data envelopment analysis · Equity · Emergency medical service

✉ M. E. Bruni
mariaelena.bruni@unical.it

S. Khodaparasti
sa.khodaparasti@sutech.ac.ir

H. R. Maleki
maleki@sutech.ac.ir

S. Jahedi
jahedi@sutech.ac.ir

P. Beraldi
patrizia.beraldi@unical.it

D. Conforti
domenico.conforti@unical.it

¹ Department of Maths, Shiraz University of Technology, Shiraz, Iran

² Department of Mechanical, Energy and Management Engineering, University of Calabria, Cosenza, Rende, Italy

1 Introduction

Major economic changes have placed increasing pressure on facility location decisions. They require large investments that can not be recovered in the short term and influence the competitive capacity of the company, affecting not only costs but also the income. For instance, considering a service business, market proximity is a critical success key to attract customers, whereas for a manufacturing business, facility location might affect procurement costs, product delivery time, and customers' service levels. In the public sector, the implications of poor location decisions are not confined to cost considerations. The effect of wrong facility location decisions may extend to disastrous consequences as increased mortality and morbidity. Thus, facility location modeling has even greater importance when applied to the siting of public, notably health care, facilities. Moreover, the public ownership of the organization and the nature of the facilities is a distinctive feature that might invalidate the use of classical facility location models. For instance, if the siting of private facilities is usually driven by the minimization of costs, the goals of public agencies, especially in health care operations, are more difficult to capture and interpret, calling for tailored approaches.

Many researchers and practitioners, since the 80s, have tried to address this issue, incorporating into the models important considerations on the equity or fairness of the service.

Mayhew and Leonardi [21] presented a model that trades off equity and efficiency, with application to health care resource allocation in London. Mandell [17] incorporated both equity and effectiveness measures into the distribution of library books in public libraries throughout a region. Cho [7] introduced an equity-efficiency trade-off model where system equity is measured by the opportunity to receive medical services, while efficiency is represented by consumer and producer welfare. More recently, Smith et al. [33] presented a range of discrete hierarchical location models with bi-criteria efficiency/equity objectives. Batta et al. [3] proposed a new location model that accounts for equity and efficiency simultaneously to demonstrate that the appropriate use of dispersion, population, and equity criteria can eliminate most of poor solutions.

The focus and contribution of this paper is the combination of equity objectives with efficient location decisions for the design of service systems in the public sector. With this aim, we present a novel model, of a general applicability, providing an integrated framework for siting facilities in an equitable and efficient manner. The proposed model captures the multi-criteria nature of the service performance, by evaluating the efficiency through the data envelopment analysis (DEA, for short). DEA is a non-parametric approach widely applied to assess the relative performance of a set of decisions making units (DMUs) that use several inputs to produce several outputs. It is worth noticing that the performance of the DMUs (the facilities in our case) is, in general, related to the allocation pattern and it can not be determined regardless of the allocation choices. To overcome this drawback, we endogenize the produced outputs, that therefore, become a function of the allocation decisions.

To the best of the authors' knowledge, this is the first study in location-allocation literature, and especially in the Emergency Medical Service (EMS) area, addressing both issues of *efficiency* and *equity* through a DEA model with endogenized outputs.

Despite the DEA has been used as a pure performance assessment methodology, only recently some attempts have been made with the aim of incorporating the evaluation of the locations into allocation decisions [26].

For instance, Thomas et al. [35] studied an obnoxious facility location problem considering both proximity and DEA scores by running the location and the DEA models iteratively. Narasimhan et al. [28] embedded branch offices efficiencies into a model, which considers capacities, demand requirements and budget constraints. Klimberg and Ratick [16] proposed two bi-objective location-allocation models by developing a simultaneous DEA model and merging it to the location model. In the proposed models, any potential allocation link is considered as a decision making unit. One of the objectives expresses the total cost of allocation while the other objective indicates the total DEA score assigned to the open links. In another research on facility location, Sahin et al. [30] presented a hierarchical model for siting blood service facilities. The results of their study show that some inefficient locations might be included in the optimal solution. The authors then suggested to improve the efficiency of optimal facilities associated with low DEA scores, reconsidering the solution of the location model on the light of the efficiency of the locations. In a recent paper, Mitropoulos et al. [26] presented a bi-objective DEA-location model in health care sector. The model includes three objectives: the minimization of total distance traveled by patients from all communities to their nearest located hospital, the minimization of underachievement in soft constraint and the maximization of the mean efficiency scores. The DEA score of facilities are computed a priori and then they are entered into the location model as parameters.

The distinctive feature of our model, with respect to the aforementioned literature, and especially with respect to the Klimberg's and Ratick's model [16], is the endogenization of the location outputs as a function of the allocation variables. In this respect, it is worthwhile noticing that the efficiency of the candidate locations is related to the allocation pattern and it can not be determined before allocating the demands. By endogenizing the outputs, we provide the decision maker with a more realistic solution which is dependent on the optimal allocation of the demand, rather than on rough estimation of future performance. Another contribution of the paper is the application of the model to a strategic EMS planning problem for locating ambulance stations, considering the randomness in the response times for the service at the demand points. This problem brings its own originality tackling the uncertainty from a risk-averse viewpoint and, therefore, enriching the EMS literature. To illustrate the applicability of the proposed model, we consider a real case study related to the strategic design of an EMS system.

The paper is organized as follows. In Sect. 2, we review the literature related to our study. In Sect. 3, the mathematical formulation of the proposed model is presented. In Sect. 4, we describe the solution approach for the proposed model. In Sect. 5, we apply the proposed model on a real case study. Finally, conclusions and discussions are drawn in Sect. 6.

2 Background on the EMS design

The EMS systems have attracted considerable attention from researchers since the 70s. The primary concern is to develop and solve mathematical models to efficiently design a system which is able to respond to emergency calls in a timely fashion.

Traditionally, emergency facility location problems dealt with the selection and dimensioning of sites and the allocation of demand points to the facility sites, therefore, many EMS models belong to the class of location-allocation models. The models proposed in the literature range from covering models, whose emphasis is on coverage within a predefined standard distance, to p -median models, which minimize the total distance over all demand points. Among the covering models, we mention the Location Set Covering Problem [34], the Maximal Covering Location Problem [8], the Maximum Expected Covering Location Problem [9], and the Maximum Availability Location Problem [29]. The majority of the models in the literature deal with the issue of efficiency by focusing on performance measures. Although the response time (RT, for short) has been traditionally considered as an adequate measure of the system performance [15], the economical pressure on health care systems highlighted that a more balanced assessment could be achieved through the use of different quantitative measures that would capture the multi-dimensional nature of emergency operations. As a result, the focus shifted to other performance measures like for example coverage and system reliability.

Aside timeliness, which is considered a critical factor directly affecting life care, different measures have been used such as cost, coverage, distance, as a proxy for the efficiency of the system [15]. To capture the relationship between different performance components and hence, to address simultaneously more than one efficiency measure, some existing researches formulated the problem as a multi-objective model [2].

The issue of equity has been only recently incorporated in the EMS literature even if this concern has a long history in the location literature [19, 22].

In order to decrease the level of disparity in the EMS location problems, Chanta et al. [4] presented three different bi-objective location models to improve the emergency service in rural areas. They classified the demands into rural and urban zones based on their geographical positions. They investigated the trade-off between efficiency and equity criteria by maximizing the expected number of covered calls on one hand, and by minimizing either the number of uncovered rural demand zones or the maximum distance between uncovered demand zones and their closest opened stations, on the other hand. Finally, they utilized the ϵ -constraint method to solve their models.

In another work, Chanta et al. [5] developed a minimum p -envy location problem to provide equitable services in emergency response location problem. To deal with the issue of equity in their model, they assigned an envy function (a function of distance) to each pair of demand nodes, for each level of priority. This value indicates the dissatisfaction level of a demand node with its serving station in comparison with other demands, for the same level of priority. The model minimizes the overall envy value. To show the validity of their model, they compared the results of the model with those obtained through the p -center and a Gini-based model, two accepted equity models in the location literature.

Following their previous work, Chanta et al. [6] presented an extension to their previous p -envy model in which the distance-based objective function is replaced with a survival function, dependent on the RT [24]. The model also imposes some lower bounds on the individual survival rates provided by the first priority servers as well as on the system-wide survival rate.

Interestingly, there are some evidences showing that the EMS planners are not very concerned about equity in services. For instance, Erkut et al. [11] highlighted the discrepancy between the concept of equity in access and the existence of different RT standards, implying different service levels in different areas, notably the rural ones.

In [36], Diaz et al. presented an equity/efficiency-based location-allocation model for large-scale EMS systems. The model considers different efficiency and equity measures to reduce disparity among both demands and servers. To account for efficiency, they applied two different criteria: the mean RT and the expected coverage. To provide an equitable service for demands, the squared coefficient of variation and the Gini coefficient were considered.

Although the reviewed literature shows the use of different equity measures, there is no concerted agreement on how equity should be measured. The choice of a set of unique evaluation metrics for equity-based models in the EMS planning is out of the scope of this paper. In the next section, we shall present a general model, irrespective of the particular equity function used.

3 Designing an equitable and efficient system

In this section, a mathematical formulation for the equity and efficiency based location-allocation problem is presented. Formally, the problem is defined on a graph with a set I of demand points. Each node $i \in I$ can be covered by a candidate location $j \in J$ in case the facility is located within a service radius τ . Let $J(i)$ be a subset of J containing the list of candidate locations that can cover node i . The aim is the optimal location of p facilities among a set J of possible candidate locations. The proposed model integrates two parts: one related to the location-allocation problem, the other concerned with the performance evaluation. For the location-allocation part of the model, we define two sets of binary variables. Variables x_j are equal to one only if the location $j \in J$ is opened, whereas variables y_{ij} refer to the allocation and are equal to one if the demand node $i \in I$ is assigned to the location $j \in J$.

In order to measure the efficiency of each candidate location j , referred in the foregoing as DMU, we use the DEA methodology. We assume that each DMU $j \in J$ has a set A of different inputs denoted by I_{aj} , $a \in A$ and a set B of outputs denoted by O_{bj} , $b \in B$. The efficiency, is defined as the ratio of the weighted sum of outputs to the weighted sum of inputs, where the weights are variables to be optimized. We denote, for each $j \in J$, by v_{aj} the variables associated with the weight of the a -th input I_{aj} and by w_{bj} the one associated with the b -th output O_{bj} . In location-allocation models, very often the performance of the locations depends on the assignments made in the design phase. Indicating by $\mathbf{y}$ the vector of the y_{ij} variables, we can define $O_{bj}(\mathbf{y})$ as the b -th endogenized output for candidate location $j \in J$. The aim is to determine the best location-allocation pattern maximizing at the same time the efficiency and the

equity of the system. We observe that equity is also a function of some measure related to the assignment pattern. Thus, we denote it as $H(\mathbf{y})$. By considering the notation introduced above, the mathematical formulation of the proposed bi-objective model can be expressed as follows:

$$\max f_1 = \sum_{j \in J} (1 - \theta_j) \quad (1)$$

$$\max f_2 = H(\mathbf{y}) \quad (2)$$

s.t.

$$\sum_{j \in J(i)} y_{ij} \leq 1 \quad \forall i \in I \quad (3)$$

$$y_{ij} \leq x_j \quad \forall i \in I, \forall j \in J \quad (4)$$

$$\sum_{j \in J} x_j \leq p \quad (5)$$

$$\sum_{a \in A} v_{aj} I_{aj} = x_j \quad \forall j \in J \quad (6)$$

$$\sum_{b \in B} w_{bj} O_{bj}(\mathbf{y}) + \theta_j = 1 \quad \forall j \in J \quad (7)$$

$$\sum_{b \in B} w_{bj} O_{bj'}(\mathbf{y}) - \sum_{a \in A} v_{aj} I_{aj'} \leq 0, \quad \forall j, j' \in J, \quad j \neq j' \quad (8)$$

$$v_{aj} \geq \delta x_j \quad \forall a \in A, j \in J \quad (9)$$

$$w_{bj} \geq \delta x_j \quad \forall b \in B, j \in J \quad (10)$$

$$y_{ij} \in \{0, 1\} \quad \forall i \in I, \quad \forall j \in J \quad (11)$$

$$x_j \in \{0, 1\} \quad \forall j \in J \quad (12)$$

$$\theta_j \geq 0 \quad \forall j \in J \quad (13)$$

The objective function (1) maximizes the total efficiency scores assigned to facilities, whereas the objective function (2) maximizes the system-wide equity performance $H(\mathbf{y})$. Constraint (3) ensures that each demand point should be assigned to at most one facility. Constraints (4) allow assignments only to open facilities. Restriction (5) imposes a limit on the number of sited facilities. Constraints (7–11) are related to the DEA part of the model. Constraint (6) states that the sum of weighted inputs for any sited facility should be equal to 1. Restrictions (7) define the inefficiency variable θ_j as a function of the weighted outputs. The set of constraints in (6) in combination with restrictions (7) and (8) ensure that the efficiency assigned with closed facilities is zero. Note that the outputs are dependent on the assignment vector $\mathbf{y}$. Constraints (8) ensure, for open facilities, that the weights are assigned in a way that the efficiency of any other facility is not greater than 1, if it uses the same weights. Moreover, constraints (9, 10), where δ is a non-Archimedean number, guarantee that only non-dominated efficient solutions in the DEA model are investigated. Finally, restrictions (11–13) define the nature of decision variables. Our model resembles the Klimberg's and Rat-

ick's [16] model, even though some distinguishing features can be highlighted. Firstly, our model considers each facility location as a potential DMU, whereas their model consider each assignment variable in a transportation network as a potential DMU. Secondly, the outputs in their model are not allowed to change and they are considered to be fixed whereas in our model the outputs are endogenized. Thirdly, the objective function has its own originality and unlike the Klimberg's and Ratick's model, does account for equity. Moreover, in order to deal with the endogeneization of the outputs, we modified the constraints accordingly. These distinctive features characterize the model, completely different in many aspects from previous works, and that can be considered as a meaningful research extension.

3.1 The EMS case

In this section, we detail the general model for determining the optimal location of the EMS facilities so as to assure a given quality of service. We consider each potential ambulance station as a DMU which uses two inputs and produces two outputs. The inputs are the capacity, measured in terms of maximum number of ambulances which can be sited at the station, and the operational cost. The outputs are related to the RT and the coverage that each station can guarantee.

Since the problem is not concerned with the more tactical problem of the EMS station dimensioning, we consider the demand arising from each station as a deterministic parameter. On the other hand, the allocation is influenced by the RT, that, as well recognized in the scientific literature, is subject to potentially dangerous fluctuations.

We model the RT between point i and station j as a random variable to faithfully represent the real behavior of the system. In particular, we assume that the RT is lognormally distributed, as commonly reported in the literature [13].

Then, station j is considered to be able to cover a demand point i only if the coverage probability, that is the probability that the RT t_{ij} is within the response time threshold (RTT, for short) τ , is greater than a reliability level $1 - \alpha$, in which α is the risk level defined by the decision maker [18]. Hence, a demand point i can be covered only by stations $j \in J(i)$, where

$$J(i) = \{j | P(t_{ij} \leq \tau) \geq 1 - \alpha\}.$$

We should mention that, although this probabilistic performance target (corresponding to the $(1 - \alpha)$ -quantile of the RT distribution) allows to exclude the more risky situations (those for which the probability of arriving on time is less than $1 - \alpha$), it is not able to exclude, even for the covered nodes the possibility that the service is provided too late. It is generally useful, in order to predict performance measures that are more closely related to patient outcomes [11, 13, 24], to consider the tail of the RT distribution to capture the delay of the RT with respect to the target.

Instead of considering a risk-neutral perspective leading to the use of the expected RT, we explicitly introduce the risk of a late answer to the emergency call. For a given random variable t_{ij} , the conditional expectation of the RT with respect to the threshold

τ can be written as $\hat{t}_{ij} = E[t_{ij}|t_{ij} > \tau] = \int_{\tau}^{\infty} t_{ij} f_{ij}(t) dt$, where f_{ij} is the probability density function of t_{ij} and E represents the expected value. This measure gives us useful information by focusing on the mean of worst case realizations, given that the threshold is violated.

For each DMU $j \in J$, we denote by O_{1j} the first output of our model representing the total expected response time, conditioned on the fact that it is greater than τ :

$$O_{1j}(\mathbf{y}) = \sum_{i \in I} \hat{t}_{ij} y_{ij}, \quad (o1).$$

It is notable that this output is an undesirable one. To deal with this case, we have regarded undesirable outputs as inputs [31]. Other methods can be used to incorporate undesirable outputs into the DEA model [14].

The second output is related to the expected demands covered by each station j which is denoted by:

$$O_{2j}(\mathbf{y}) = \sum_{i \in I} p_{ij} d_i y_{ij}, \quad (o2),$$

where p_{ij} is the probability that the RT is within the threshold τ and d_i is the demand associated with the demand point i .

To address the issue of equity in our model, we use the equity criterion of minimizing the total number of uncovered demand zones. Chanta et al. [4] used this criterion as a measure of fairness for the ambulance location problem in a tactical level.

The EMS location-allocation model balancing equity and efficiency, which hereinafter is called E^3MS , for the sake of brevity, is then defined by problem (1–13) amended with constraints (o1) and (o2), where $H(\mathbf{y}) = \sum_{i \in I} [\sum_{j \in J(i)} y_{ij}]$.

4 The exact solution approach

The proposed model is very difficult to deal with. It is a bi-objective mixed integer non-linear problem. In order to address the non-linearity, we adopt a standard linearization technique to transform the model into a bi-objective mixed integer problem. In particular, we can apply linear McCormick [23] envelopes to linearize the non-linear terms in constraints (7, 8) resulting from the product of continuous and binary variables.

Defining the auxiliary variable z_{bij} , as $z_{bij} = w_{bj} y_{ij}$, we can come up with the following set of linear constraints

$$z_{bij} \geq w_b^{max} y_{ij} + w_{bj} - w_b^{max}, \quad \forall b \in B, \quad \forall i \in I, \quad \forall j \in J \quad (14)$$

$$z_{bij} \geq 0, \quad \forall b \in B, \quad \forall i \in I, \quad \forall j \in J \quad (15)$$

$$z_{bij} \leq w_b^{max} y_{ij}, \quad \forall b \in B, \quad \forall i \in I, \quad \forall j \in J \quad (16)$$

$$z_{bij} \leq w_{bj}, \quad \forall b \in B, \quad \forall i \in I, \quad \forall j \in J \quad (17)$$

where w_b^{max} is an upper bound of the output weight w_{bj} .

The remaining part of this section is devoted to the presentation of an exact method for our bi-objective model.

Different solution approaches have been proposed in the literature for computing the Pareto front for the class of mixed integer bi-objective problems, among which the weighting method is the most popular one. The method scalarizes the objective function vector, reducing the problem to a single objective one. The method exhibits several drawbacks, for example, the dependence of the performance on the weighting coefficients and the lack of guarantee that the optimal Pareto front is obtained. Another well-known and widely used method is the so called ϵ -constraint method which consists in the repeated solution of a sequence of single objective problems, amended with additional constraints. In particular, one objective is chosen to be optimized and the other objective is incorporated as an inequality constraint where the right hand side (the parameter ϵ) assumes different values in a certain range. Since the method can also find weak efficient solutions, a modification of ϵ -constraint, specifically tailored for multi-objective problems, has been presented in [20] by Mavrotas.

Our model is a bi-objective mixed integer optimization problem, where one objective consists in the maximization of the total number of covered demand points and the other in the maximization of the total system efficiency. By choosing the efficiency criterion as the main objective function and considering the equity criterion as a constraint, the ϵ -constraint problem $P(\epsilon)$ is defined as

$$\begin{aligned} \max \quad & f_1(X) \\ \text{s.t.} \quad & f_2(X) \geq \epsilon \quad P(\epsilon) \\ & X \in \mathcal{X} \end{aligned}$$

where X denotes the set of variables defined in the mathematical model of Sect. 3 plus the auxiliary linearization variables introduced above. In addition, $\mathcal{X}$ is the set of constraints in (3–13), (o1), and (o2).

The idea is to construct a sequence of ϵ -constraint problems based on a progressive reduction of ϵ . The set of values that ϵ may take is finite since the first objective represents the number of covered demands which is finite and integer.

In order to define the range of variations for ϵ , we first solve the E^3MS problem considering only the objective function related to the equity, to find the ideal point $f_2^I = \max_{X \in \mathcal{X}} f_2(X)$. Then, we optimize the efficiency related objective function alone to obtain the ideal point $f_1^I = \max_{X \in \mathcal{X}} f_1(X)$. Next, we apply the lexicographic optimization to obtain a Pareto optimal (non-dominated) solution. In practice, the lexicographic optimization is performed by solving the following problems

$$\begin{aligned} f_1^N &= \max_{X \in \mathcal{X}} f_1(X) \\ \text{s.t.} \quad & f_2(X) = f_2^I \end{aligned}$$

and

$$f_2^N = \max_{X \in \mathcal{X}} f_2(X)$$

$$s.t. f_1(X) = f_1^I$$

The points f_1^N and f_2^N are called the nadir values.

After this calculation, we obtain, for the second objective function, a discrete set of integer points in $[f_2^N, f_2^I]$, whose values are iteratively assigned to ϵ . Let the cardinality of this range be C and ϵ^c be the element in the c -th position in the range. The scheme of the algorithm is reported in Algorithm 1.

Algorithm 1 The exact ϵ -constraint method

Let EFF be the set of generated efficient solutions.

$EFF = \emptyset$.

for $c = C, \dots, 1$ **do**

Solve $P(\epsilon^c)$ and let $f_1^*(\epsilon^c)$ and $S^*(\epsilon^c)$ be the optimal objective function value and the optimal solution, respectively.

if $c <> C$ **then**

if $f_1^*(\epsilon^c) <> f_1^*(\epsilon^{c-1})$ **then**

$EFF = EFF \cup S^*(\epsilon^c)$.

end if

end if

end for

return EFF .

Lemma 1 The point (f_1^N, f_2^I) is an efficient solution.

Proof Suppose that the point (f_1^N, f_2^I) is not efficient. Therefore, it should exist another point (z_1, z_2) such that either:

1. $z_1 = f_1^N$ and $z_2 > f_2^I$
2. $z_1 > f_1^N$ and $z_2 > f_2^I$
3. $z_1 > f_1^N$ and $z_2 = f_2^I$

Cases (1) and (2) are in contradiction with the fact that f_2^I is an ideal point. The case (3) is in contrast with the fact that f_1^N is a nadir point. Therefore, the point (f_1^N, f_2^I) is an efficient point. Similarly, we can prove that the point (f_1^I, f_2^N) is also an efficient solution. $\square$

Theorem 1 The set of solutions produced by the Algorithm 1 generates the exact Pareto front of our model.

Proof In order to proof the claim, we should first prove that each solution belonging to the set EFF is Pareto optimal and second, we should prove that the set of solutions produced by Algorithm 1 gives the exact Pareto front.

In order to prove the first part, let us consider a solution $S \in EFF$. S is the optimal solution of a problem $P(\epsilon^c)$ for some $c = C, \dots, 1$. We might have two different

cases: either $f_2(S) = \epsilon^c$ or $f_2(S) > \epsilon^c$. If $f_2(S) = \epsilon^c$, S is a Pareto optimal solution, as shown for the general case (see [25]).

On the other hand, if $f_2(S) > \epsilon^c$, since f_2 involves only binary variables and ϵ^c is integer, $f_2(S) = \epsilon^f$, with $f > c$ and therefore, S is the optimal solution of a problem solved before problem $P(\epsilon^c)$.

To prove the second part, we should show that any other solution $S \notin EFF$ is not efficient. Assume that there is a solution S for the E^3MS problem which does not belong to the efficient set EFF . Since S is feasible, $f_2(S)$ assumes an integer value. Let $f_2(S) = \epsilon^c$ for some $c = C, \dots, 1$ and therefore, it is an efficient solution, which contradicts the hypothesis. $\square$

5 The computational results

In this section, we report on the results collected by applying the proposed model on a real case study for strategic EMS design in Edmonton city, Canada.

The information about the case study and data set is based on the previous researches carried out by Erkut et al. [11, 13] and Erdogan et al. [12]. The data set includes the average and the standard deviation of lognormally distributed response times, the average number of calls, and the capacity of stations for a set of 180 demand points and 16 potential ambulance stations. The interested reader can find the data on the following link: “<http://professor.business.ualberta.ca/armanningolfsson/Research/Data>”.

Since the aforementioned data set does not include the cost of siting stations, we generated them from the interval [10–32].

To evaluate the performance of solutions, concerning the fairness criterion, we apply the Gini coefficient of the Lorenz curve, the within-group Gini indices associated with covered ($Gini_1^w$) and uncovered ($Gini_0^w$) zones, and the gross between-group index (G^{gb}) in Dagum's Gini decomposition.

It is notable that the within-group Gini indices $Gini_1^w$ and $Gini_0^w$ quantifies the inequality among each individual group of covered and uncovered zones, respectively while the gross within-group Gini index captures the inequalities between the set of covered and uncovered demand zones.

Based on Dagum's Gini decomposition, the Gini index is decomposed as follows [27]:

$$\begin{aligned} Gini &= Gini_0^w + Gini_1^w + Gini^{gb} \\ Gini_0^w &= \frac{\sum_{i \in N_0} \sum_{k \in N_0} |\sum_{j \in J} t_{ij} y_{ij} - \sum_{j \in J} t_{kj} y_{kj}|}{2|I| \sum_{i \in I} \sum_{j \in J} t_{ij} y_{ij}} \\ Gini_1^w &= \frac{\sum_{i \in N_1} \sum_{k \in N_1} |\sum_{j \in J} t_{ij} y_{ij} - \sum_{j \in J} t_{kj} y_{kj}|}{2|I| \sum_{i \in I} \sum_{j \in J} t_{ij} y_{ij}} \\ Gini^{gb} &= \frac{\sum_{i \in N_1} \sum_{k \in N_0} |\sum_{j \in J} t_{ij} y_{ij} - \sum_{j \in J} t_{kj} y_{kj}|}{|I| \sum_{i \in I} \sum_{j \in J} t_{ij} y_{ij}} \end{aligned}$$

where $N_0 = \{i \in I | \sum_{j \in J} y_{ij} = 0\}$ and $N_1 = \{i \in I | \sum_{j \in J} y_{ij} = 1\}$.

To capture the randomness in our model, we define the Gini index based on the quantile of lognormally distributed RT by replacing the distance values in the traditional Gini index with the quantile of RTs [10]. In addition, we classify demand points into two separate groups of covered and uncovered zones. This enables us to investigate the inequality among both covered and uncovered zones. Although, due to the spatial configuration of potential stations, some demand zones are not covered within the RTT, in general and in the absence of auxiliary servers, the closest open station would respond to them. As a result, we assign the uncovered zones to their closest open station enabling us to capture the inequality among uncovered zones as well as between covered and uncovered zones.

The algorithm and problem modeling have been implemented in AIMMS 4.1 [38] and solved by ILOG CPLEX 12.6 [37]. The experiments were executed on a laptop Intel core i7 with a 2.7 GHz processor and 4 GB RAM. The average solution time is around 900 seconds.

The first set of computational experiments is devoted to the analysis of the behavior of the model as a function of different risk levels α and RTTs.

Figure 1 reports the trade-off between the risk level α and the equity components of the Gini coefficient discussed above for the single objective problem optimizing f_2 . The instance solved has $p = 12$ and the RTT equal to 10 minutes. The results show that lower values for equity measures of $Gini_0^w$, $Gini^{gb}$ are obtained for higher risk levels. This can be explained by considering that, the higher is the risk level that the

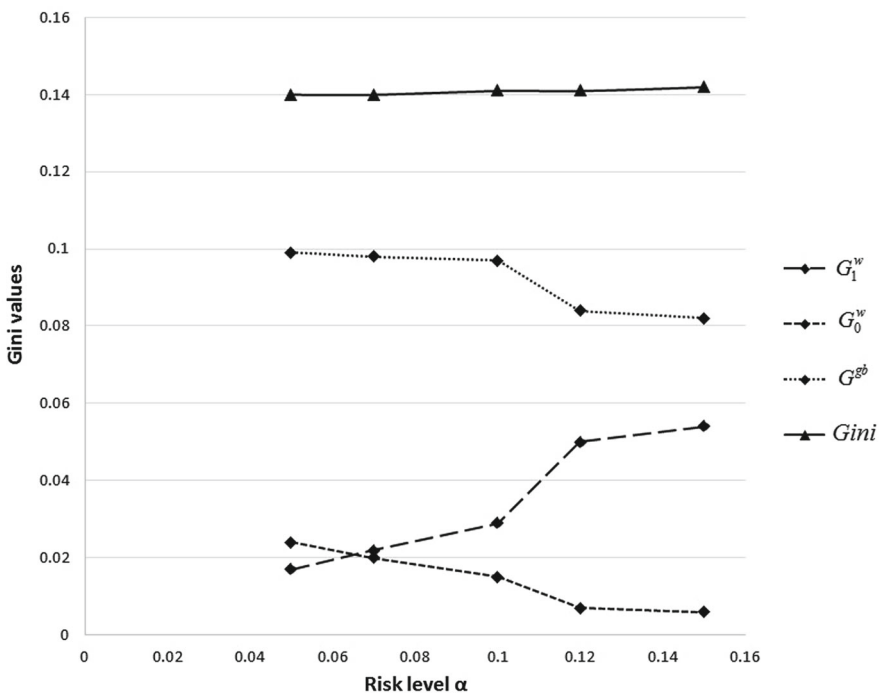


Fig. 1 Risk level α vs. within- and between-group Gini indices

Table 1 E^3MS in comparison with pure efficiency and equity models

Optimization criterion	f_1	f_2
f_1	12	80
f_2	10.67	98
$f_1 f_2 = f_2^I$	11.299	98
$f_2 f_1 = f_1^I$	12	91

decision maker is willing to bear, the larger is the number of covered demand points. For instance, by varying the risk level from 0.05 to 0.15, the total number of covered zones increases from 98 to 141. This, in turn, results in a more balanced system configuration (lower values for $Gini^{sb}$), where equality in the covered demands is sacrificed to achieve better service for the uncovered zones. The decrease in the within-group Gini coefficient $Gini_0^w$ and, on the contrary, the increase in the $Gini_1^w$ support this claim. We would like to remark that the total Gini index will increase slightly, indicating that the decrease in the reliability level $1 - \alpha$ (the increase in the risk level α) will not necessarily produce more equitable solutions.

A note of caution should be in order. The results show a high sensitivity of the number of covered zones to the variations of the risk level. As an example, a 5 %-increase in the risk level (from 0.05 to 0.1) will increase the number of covered demands by 18 % while, by increasing α from 0.1 to 0.15, about 22 % of uncovered zones can be covered. On the other hand, there is a dependence between the risk level and the RTT. By increasing the RTT from 10 to 15 minutes, the number of covered demands will increase to a different extent, considering different risk levels. For α equal to 0.05, 0.10, and 0.15, we obtain an increase of about 70, 49, and 24 %, respectively, in the number of covered zones. This highlights the importance of choosing the appropriate values for the risk level and the RTT.

The second set of experiments is devoted to the validation of the proposed model, with respect to the pure equity- and pure efficiency-based models.

With this aim, we solve the E^3MS model, considering separately the two single objectives f_1 and f_2 considering $\alpha = 0.05$, $RTT = 10$ minutes and $p = 12$. The results have been reported in the Table 1, where the headers have the following meaning. The first column represents the criterion to be optimized, the second and third columns are the objective functions evaluated in the optimal solutions obtained. For the sake of clarity, we recall that the first two rows of the Table are related to the ideal points discussed in Sect. 3, whereas the last two rows are equivalent to the lexicographic optimization, and the corresponding solutions are the nadir points. As we can see, the nadir point f_1^N outperforms the ideal point f_2^I with respect to the efficiency criterion. This reveals the fact that applying the combined E^3MS model enables the decision maker to improve the efficiency criterion while the highest equity level is guaranteed. The resulting configuration of the open stations as well as the assignment pattern exhibits the best performance for the equity criterion while producing more outputs and therefore increasing the overall efficiency. In a similar way, the nadir point f_2^N outperforms the ideal point f_1^I in the equity criterion. In this case, we are able to provide higher values for the equity measure even when the overall efficiency is maximum.

Table 2 The results for the Pareto front with $\alpha = 0.05$ and $RTT = 10$ minutes

Solution	f_1	f_2	$Gini_1^w$	$Gini_0^w$	$Gini^{8b}$	$Gini$	SEC (%)	SCRT
1	11.327	97	0.017	0.025	0.098	0.140	91.90	307.21
2	11.357	96	0.017	0.026	0.097	0.140	92.80	307.29
3	11.392	95	0.016	0.027	0.097	0.140	93.69	308.42
4	11.434	94	0.016	0.028	0.096	0.140	94.58	308.51
5	11.478	93	0.015	0.029	0.095	0.139	95.54	309.67
6	11.540	92	0.015	0.030	0.094	0.139	96.35	310.86

Table 2 reports the efficiency and equity performance measures of the Pareto optimal solutions for the case in which $\tau = 10$ minutes and $\alpha = 0.05$.

The columns 2 and 3 show the values of efficiency and equity criteria for the Pareto optimal solutions. The other columns, but the last two, represent the equity components in Dagum's decomposition as well as the Gini's coefficient. In the last two columns, we reported the percentage of the system-wide expected coverage SEC and the average RT above the threshold, SCRT in seconds.

The results in Table 2 indicate that the decrease in the number of covered demands will result in higher values for within-group Gini index assigned with uncovered zones $Gini_0^w$. In this situation, the disparity in uncovered zones is increased at the price of improving the efficiency. This agrees with our concept of equity, since the emphasis is on the uncovered demands. Therefore, it is reasonable that the equality in the uncovered group is higher, when the equity function is maximized. Higher values for the within-group Gini index for covered demands $Gini_1^w$, when our equity measure is maximized (solution 1), is an evidence of the fact that our equality measure is focused on the uncovered zones. In fact, the access to the service can be extended to cover more points, at the expense of a slight decrease in equality in covered demands. This shows also the trade-off between horizontal and vertical equity. Horizontal equity implies that all the demand points are considered in an equal manner. It means that suburban nodes are treated without discrimination. Vertical equity is concerned with the distribution of the service between different groups. By this definition, EMSs are equitable if they favor disadvantaged groups, compensating for overall inequities. In our case, horizontal equity is not in contrast with vertical equity, since, when suburban points are disadvantaged in terms of equity in access to the service (solution 6), the disparity between covered and uncovered zones reduces. This implies a decrease in the disparity between covered and uncovered zones. As a last remark, we should remark that the total Gini index exhibits only a slight variation over the Pareto front. This verifies that, although our main equity criterion is not explicitly known as a RT-based equity measure, it is not in contrast with the results of Gini performance measure.

The results in the last two columns show the trade-off between two outputs in the DEA model. It is evident that sacrificing the system equity will lead to better results for the system-wide expected coverage performance. In particular, the system-wide outputs of SEC and SCRT improve when the efficiency in the system is maximized. We should notice, as a shortcoming of our model, that the choice of the weights in the

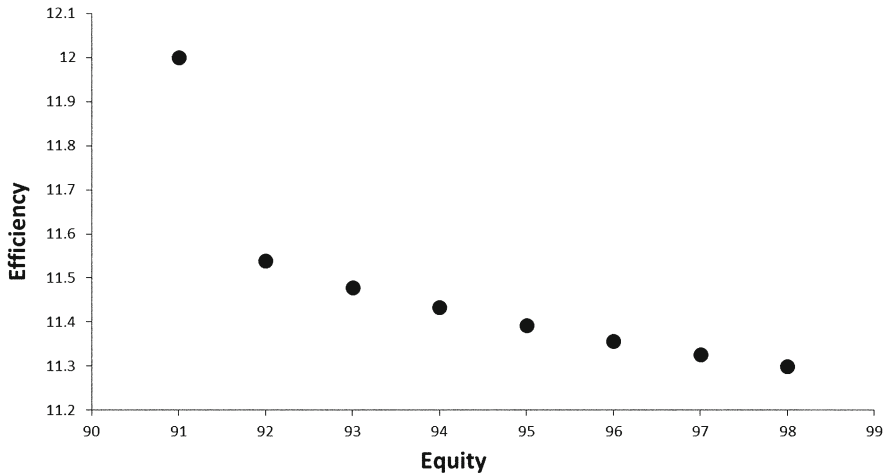


Fig. 2 The trade-off between efficiency and equity criteria in Pareto front

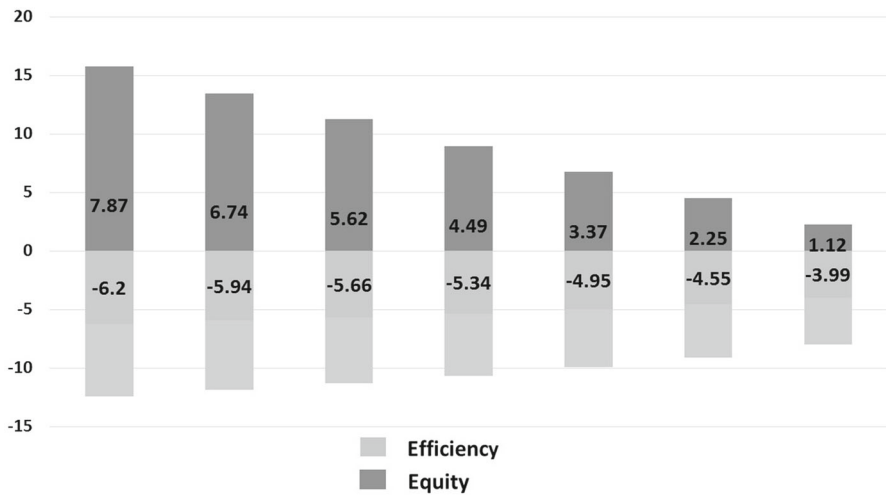


Fig. 3 The percentage of trade-off: efficiency versus equity criteria in Pareto front

DEA model, and consequently the outputs, is out of the decision maker's control, and all the outputs are considered equal.

Figure 2 represents the exact Pareto front for the case in which $\alpha = 0.05$ and $\tau = 10$ minutes.

The trade-off between equity and efficiency for the Pareto efficient solutions is shown in Fig. 3.

The last Pareto efficient solution in the Pareto front, which has the best value for the efficiency criterion, is considered as the baseline of this comparison. The positive values show the percentage of improvement in the equity criterion while the negative values indicate the decrease in the efficiency criterion. This provides the decision

maker a framework to find that how much loss should undertake in one criterion for the benefit of a certain improvement in another criterion.

6 Conclusion

The most significant contributions of the paper can be summarized as follows. In this paper, an integrated location-allocation model for balancing equity and efficiency has been proposed. We address more than one efficiency performance measure simultaneously, by using a DEA model, while existing researches formulate the problem as a multi-objective model.

By endogenizing the outputs of the DEA, a better trade-off between equity and efficiency criteria is made. This issue is highly important since in location-allocation problems, the output of each DMU is dependent on the assignment variables. It is obvious that before siting facilities, we can not estimate the output parameters.

In addition, an application of the model to the strategic design of EMS has been presented. The model has its own contribution in the EMS literature, since captures the uncertainty in RT in a novel way through the incorporation of a risk averse approach. In this applicative context, an interesting issue would be the exploration of the trade-off between equity and efficiency also in tactical-operational models, as for instance in location-relocation models (see [1, 32] and the references therein), where a stable performance should be guaranteed.

As already noticed, the choice of weights in the DEA model, and consequently the outputs, is out of the decision maker's control. In some cases, in which the decision maker has some priority levels on the outputs, this may probably result in undesirable results. To overcome this drawback, one suggestion can be the incorporation of weights restrictions into DEA models.

As another limitation of the proposed model, we can refer to the need for developing a tailored solution method which considers the peculiarity of the class of location-allocation models. This can be a motivation for future researches to explore the structure of the problem in more detail.

Finally, we should mention that although a general form for the equity function has been considered, there is an essential need for the investigation of equity measures designed especially for EMS systems. Future research will also focus on the application of the proposed model to other interesting applicative contexts in the health sector and in other fields.

References

1. Alanis, R., Ingolfsson, A., Kolfal, B.: A Markov chain model for an EMS system with repositioning. *Prod. Oper. Manage.* **22**(1), 216–231 (2013)
2. Araz, C., Selim, H., Ozkarahan, I.: Fuzzy multi-objective covering-based vehicle location model for emergency services. *Comp. Oper. Res.* **34**, 705–726 (2007)
3. Batta, R., Lejeune, M., Prasad, S.: Public facility location using dispersion, population, and equity criteria. *Euro. J. Oper. Res.* **234**, 819–829 (2014)
4. Chanta, S., Mayorga, M.E., McLay, L.A.: Improving emergency service in rural areas: a bi-objective covering location model for EMS systems. *Ann. Oper. Res.* **221**(1), 133–159 (2011)

5. Chanta, S., Mayorga, M.E., Kurz, M.E., McLay, L.A.: The minimum p-envy location problem: a new model for equitable distribution of emergency resources. *IIE Trans. Healthcare Syst. Eng.* **1**(2), 101–115 (2011)
6. Chanta, S., Mayorga, M.E., McLay, L.A.: The minimum p-envy location problem with requirement on minimum survival rate. *Comp. Ind. Eng.* **74**, 228–239 (2014)
7. Cho, C.J.: An equity-efficiency trade-off model for the optimum location of medical care facilities. *Soc.-Eco. Plan. Sci.* **32**(2), 99–112 (1998)
8. Church, R.L., ReVelle, C.: The maximal covering location problem. *Papers Reg. Sci. Assoc.* **32**, 101–118 (1974)
9. Daskin, M.S.: A maximal expected covering location model: formulation, properties, and heuristic solution. *Trans. Sci.* **17**, 48–69 (1983)
10. Drezner, T., Drezner, Z., Guyse, J.: Equitable service by a facility: Minimizing the Gini coefficient. *Comp. Oper. Res.* **36**, 3240–3246 (2009)
11. Erkut, E., Ingolfsson, A., Erdogan, G.: Ambulance location for maximum survival. *Naval Res. Log.* **55**(1), 42–58 (2008)
12. Erdogan, G., Erkut, E., Ingolfsson, A., Laporte, G.: Scheduling ambulance crews for maximum coverage. *J. Oper. Res. Soc.* **61**(4), 543–550 (2009)
13. Erkut, E., Ingolfsson, A., Sim, T., Erdogan, G.: Computational comparison of five maximal covering models for locating ambulances. *Geograph. Anal.* **41**(1), 43–65 (2009)
14. Fare, R., Grosskopf, S.: Modeling undesirable factors in efficiency evaluation: Comment. *Euro. J. Oper. Res.* **157**(1), 242–245 (2004)
15. Goldberg, J.: Operations Research Models for the Deployment of Emergency Services Vehicles. *EMS Manag. J.* **1**, 20–39 (2004)
16. Klimberg, R.K., Ratick, S.J.: Modeling data envelopment analysis (DEA) efficient location/allocation decisions. *Comp. Oper. Res.* **35**, 457–474 (2008)
17. Mandell, M.: Modeling effectiveness-equity tradeoffs in public service delivery systems. *Manag. Sci.* **37**(4), 467–482 (1991)
18. Marianov, V., ReVelle, C.: The queueing maximal availability location problem: a model for the siting of emergency vehicles. *Euro. J. Oper. Res.* **93**(1), 110–120 (1996)
19. Marsh, M.T., Schilling, D.A.: Equity measurement in facility location analysis: A review and framework. *Euro. J. Oper. Res.* **74**(1), 1–17 (1994)
20. Mavrotas, G.: Effective implementation of the e-constraint method in Multi-Objective Mathematical Programming problems. *Appl. Math. Comp.* **213**, 455–465 (2009)
21. Mayhew, L.D., Leonardi, G.: Equity, efficiency, and accessibility in urban and regional health-care systems. *Environ. Plan. A* **14**(11), 1479–1507 (1982)
22. McAllister, D.M.: Equity and efficiency in public facility location. *Geograph. Anal.* **8**(1), 47–63 (1976)
23. McCormick, G.P.: Computability of global solutions to factorable nonconvex programs: Part I Convex underestimating problems. *Math Program.* **10**, 147–175 (1976)
24. McLay, L.A., Mayorga, M.E.: Evaluating emergency medical service performance measures. *Health Care Manag. Sci.* **13**, 124–136 (2010)
25. Miettinen, K.M.: *Nonlinear Multiobj. Optimiz.* Kluwer, Boston (1998)
26. Mitropoulos, P., Mitropoulos, I., Giannikos, I.: Combining DEA with location analysis for the effective consolidation of services in the health sector. *Comp. Oper. Res.* **40**(9), 2241–2250 (2013)
27. Mussard, S., Alperin, M. N. P., Seyte, F., Terraza, M.: Extensions of Dagum's Gini decomposition (No. 05–07). Working Paper 05–07, Dpartement d'économie de la Faculté d'administration l'Université de Sherbrooke (2005)
28. Narasimhan, R., Talluri, S., Sarkis, J., Ross, A.: Efficient service location design in government services: a decision support system framework. *J. Oper. Manag.* **23**, 163–178 (2005)
29. ReVelle, C., Hogan, K.: The maximum availability location problem. *Trans. Sci.* **23**, 192–200 (1989)
30. Sahin, G., Sural, H., Meral, S.: Locational analysis for regionalization of Turkish Red Crescent blood services. *Comp. Oper. Res.* **34**, 692–704 (2007)
31. Scheel, H.: Undesirable outputs in efficiency evaluation. *Euro. J. Oper. Res.* **132**, 400–410 (2001)
32. Schmid, V.: Solving the dynamic ambulance relocation and dispatching problem using approximate dynamic programming. *Euro. J. Oper. Res.* **219**(3), 611–621 (2012)
33. Smith, H.K., Harper, P.R., Potts, C.N.: Bicriteria efficiency/equity hierarchical location models for public service application. *J. Oper. Res. Soc.* **64**(4), 500–512 (2013)

34. Toregas, C.R., Swain, R., ReVelle, C.S., Bergman, L.: The location of emergency service facilities. *Oper. Res.* **19**, 1363–1373 (1971)
35. Thomas, P., Chan, Y., Lehmkuhl, L., Nixon, W.: Obnoxious-facility location and data-envelopment analysis: A combined distance-based formulation. *Euro. J. Oper. Res.* **141**(3), 495–514 (2002)
36. Toro-Diaz, H., Mayorga, M. E., McLay, L. A., Rajagopalan, H. K., Saydam, C.: Reducing disparities in large-scale emergency medical service systems. *Journal of the Operational Research Society.* 1–13 (2014)
37. CPLEX. ILOG, C.P.L.E.X.: 6.5: Users manual. CPLEX Optimization Inc, Incline Village, NV (1999)
38. Bisschop, J., Roelofs, M.: AIMMS. 3.7 Users guide. Paragon Decision Technology B.V., The Netherlands (2006)